

LECTURE 1 (SEPTEMBER 2ND)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C . Then, the contour integral of f is defined as

$$\int_C f(z) dz = \int_C u dx - v dy + i \int_C u dy + v dx$$

and also, through parametrisation, we have

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connected bounded region whose boundary is C . Then,

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where C is oriented in the positive sense. Here,

$$\int_C P dx = - \iint_D \frac{\partial Q}{\partial y} dy dx$$

and

$$\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dx dy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P dx = \int_a^b P(x, g_1(x)) dx - \int_a^b P(x, g_2(x)) dx$$

and

$$\iint_D \frac{\partial P}{\partial y} dy dx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b \left[P(x, g_2(x)) - P(x, g_1(x)) \right] dx$$

Similarly,

$$\int_D Q \, dy = \int_c^d Q(y, h_2(x)) \, dy - \int_c^d Q(h_1(y), y) \, dy$$

and

$$\iint \frac{\partial Q}{\partial x} \, dx \, dy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x} \, dx \, dy = \left[Q(h_2(y), y) - Q(h_1(y), y) \right] dy$$

□

Theorem. (Cauchy's theorem) If f is analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z) \, dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D . Thus

$$\int_C f(z) \, dz = \int_C u \, dx - v \, dy + i \left[\int_C u \, dy + v \, dx \right]$$

where

$$\int_C u \, dx - v \, dy = \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

and

$$\int_C u \, dy + v \, dx = \iint_D \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z) \, dz = 0$$

where ∂D is a positively oriented boundary of D .

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} \, dz$$

LECTURE 2 (SEPTEMBER 4TH)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C , we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi$$

for all z inside C . If $\bar{D}(z_0, r) = \{z \mid |z - z_0| \leq r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0) \left(1 - \frac{z - z_0}{\xi - z_0}\right)}$$

for $z \in D(z_0, r)$. However,

$$\left| \frac{z - z_0}{\xi - z_0} \right| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_C f(\xi) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}} d\xi \\ &= \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi = \frac{1}{n!} f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \rightarrow f$ uniformly on C

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} c_n r^n e^{int} \sum_{m=0}^{\infty} \bar{c}_m r^m e^{-imt} dt \\
&= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{2\pi} \int_0^{2\pi} c_n \bar{c}_m r^{n+m} e^{i(n-m)t} dt \\
&= \sum_{n=0}^{\infty} |c_n|^2 r^{2n}
\end{aligned}$$

Here, we note that

$$\int_0^{2\pi} e^{i(n-m)t} dt = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases}$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)| \leq M$ for all $z \in \mathbf{C}$ then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq M^2$$

for all $r > 0$ which means that $c_1 = c_2 = \dots = 0$.

□

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n \rightarrow z_0 \in D$. If f is analytic in D with $f(z_n) = 0$ for all $n \in \mathbf{N}$, then $f(z) = 0$ for all $z \in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D . Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$|f(z_0 + re^{it})| \leq |f(z_0)|$$

for all $t \in [0, 2\pi]$, then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq |f(0)|^2 = |c_0|^2$$

which means that $c_1 = \dots = c_n = 0$. Then we can apply the identity theorem.

□

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z) dz = 0$$

for all closed contours C in D , then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi) d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z . Then

$$F(z) - F(z_0) = \int_{[z_0, z]} f(\xi) d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0, z]} [f(\xi) - f(z_0)] d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \frac{1}{|z - z_0|} |z - z_0| \varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then f is analytic on D .

Example. On $\mathbf{R}$, consider

$$f_n(x) = \sum_{k=1}^n \frac{\cos 3^k x}{2^k} \rightarrow f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D . Then

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz = 0$$

By Morera's theorem, the proof is complete. $\square$

LECTURE 3 (SEPTEMBER 9TH)

Theorem. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

- (i) There is g analytic on D such that $e^g = f$ on D
- (ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z) dz = 0$ for every closed contour C in D . Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^z f(\xi) d\xi$$

is independent of path. Since f has no zero in D , f'/f is analytic on D . We define g on D by

$$g(z) = \int_{z_0}^z \frac{f'(\xi)}{f(\xi)} d\xi + \text{Log } f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D . Observe that

$$\begin{aligned} (fe^{-g})' &= f'e^{-g} + f(e^{-g})' \\ &= f'e^{-g} + fe^{-g}\left(-\frac{f'}{f}\right) \\ &= f'e^{-g} - f'e^{-g} = 0 \end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then

$$\begin{aligned} h(z_0) &= f(z_0)e^{-g(z_0)} \\ &= f(z_0)e^{-\text{Log } f(z_0)} \\ &= f(z_0)[f(z_0)]^{-1} = 1 \end{aligned}$$

(2) By (1), there exists an analytic g such that $f = e^g$ on D . If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Riemann mapping theorem) If D is a simply connected domain in $\mathbf{C}$ ($D \neq \mathbf{C}$), then there is a one-to-one analytic function f from D onto $U = \{z \in \mathbf{C} \mid |z| < 1\}$ (the unit disk).

Proof. The proof consists of three steps. Let Σ be a set of all 1-1 analytic functions from D to U .

$$\Sigma = \{f : D \rightarrow U \mid f \text{ is 1-1 analytic}\}$$

We first prove that

- (i) $\Sigma \neq \emptyset$
- (ii) That if $\psi \in \Sigma$ satisfies $\psi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$,
- (iii) For a given $z_0 \in D$ if $\eta_0 = \sup\{|\psi'(z_0)| \mid \psi \in \Sigma\}$ then there is $f \in \Sigma$ such that $|f'(z_0)| = \eta_0$.

□

Theorem. (Exhaustion of D by compact subsets) Let $D \subset \mathbf{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

- (i)

$$K_1 \subset K_2 \subset \dots \subset K_n \subset D$$

- (ii)

$$\bigcup_{n=1}^{\infty} K_n = D$$

- (iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \mathbf{N}$

- (iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that

$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$

for each $z \in K_n$

Proof. Define

$$K_n = \left\{z \in \mathbf{C} \mid |z| \leq n\right\} \cap \left\{z \in \mathbf{C} \mid d(z, D^c) \geq \frac{1}{n}\right\}$$

$$G_n = \left\{|z| < n\right\} \cap \left\{d(z, D^c) > \frac{1}{n}\right\}$$

and $G_n = D$ by the Archimedean property.

□